

An Improved Repetitive Control Scheme for Grid-Connected Inverter With Frequency-Adaptive Capability

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Abstract—The power quality of grid-connected inverters has drawn a lot of attention with the increased application of distributed power generation systems. The repetitive control technique is widely adopted in these systems, due to its excellent tracking performance and low output total harmonic distortion (THD). However, in an actual system, the ratio of the sampling frequency to the grid frequency cannot always maintain an integer, and then, the resonant frequencies of the repetitive control technique will deviate from the real grid fundamental and harmonic frequencies. This will degrade the performance of the system, particularly when the grid frequency varies. Even if the ratio is a fixed integer, the auxiliary function for stabilization in the conventional repetitive control technique will also increase the steady-state tracking error and THD of the system. In this paper, an improved repetitive control scheme with a special designed finite impulse response (FIR) filter is proposed. The FIR filter cascaded with a traditional delay function can approximate the ideal repetitive control function of any ratio. The proposed scheme varies the FIR filter according to varied grid frequency and maintains its resonant frequencies matching the grid fundamental and harmonic ones. Finally, the simulation and experimental results show that the improved repetitive control scheme can effectively reduce the tracking error and compensate harmonics of the inverter systems.

Index Terms—Finite impulse response (FIR) filter, frequency variation, grid-connected inverter, repetitive control, total harmonic distortion (THD).

I. INTRODUCTION

A LONG WITH the new energy and renewable energy source development, distributed power generation systems (DPGSs) have attracted a lot of interest in the latest years. Moreover, the pulsewidth modulation (PWM) voltage-source inverter (VSI) is a key part of the system. To decrease pollution to the grid, the grid-connected VSI should have high output power factor and low output current total harmonic distortion (THD). For both photovoltaic arrays and wind turbines connected to the grid, the allowed maximum current THD is 5% [1].

The most popular current control methods for grid-connected inverters are proportional–integral (PI) control and

proportional–resonant (PR) control [2]–[8]. The PI control method is usually used in the synchronous reference frame (SRF) and can work well under balanced systems. However, it is not suitable for unbalanced systems, which is common in the DPGS. The PR control method is popular in the stationary reference frame, because it can eliminate the tracking error while regulating sinusoidal signals. However, while compensating harmonics, the PR control method has to adopt one regulator for each harmonic frequency. Moreover, in order to maintain good performance, all of the resonant frequencies have to be identical to the grid fundamental and harmonic frequencies, which keep varying in the DPGS [2].

The repetitive control technique, using the internal model principle [9], can effectively eliminate output steady-state tracking error and minimize THD, as it generates high gains at the fundamental and harmonic frequencies. The internal model of an ac system can be obtained by adding a delay function into a positive feedback loop. It has been widely applied to single-phase VSIs [10]–[15], three-phase VSIs [16]–[18], and active power filters [19], [20]. Usually, in the repetitive control technique, the ratio of the sampling frequency to the grid frequency is an integer. However, this ratio cannot maintain an integer in some cases: 1) The grid frequency varies around its rated value, which is common in the DPGS, and 2) the microcontrol unit (MCU)'s clock frequency is 40 MHz, and the sampling frequency should be an integral division of it, for example, 10 kHz, while the grid frequency is 60 Hz. In these cases, the performance of the system will be degraded significantly using the conventional repetitive control technique. Meanwhile, there is an auxiliary function in the conventional repetitive control technique. This auxiliary function is cascaded with a traditional delay function and can guarantee the stabilization of the system. However, it will increase the steady-state tracking error and THD, even when the ratio is an integer.

The repetitive control method proposed in [21] and [22] uses a variable one-order low-pass filter to adapt to the varied grid frequency. Although its resonant point and the fundamental frequency can be perfectly matched, the rest of the resonant points still deviate from the harmonics. This will decrease the harmonic compensation performance. A repetitive control method with fictitious sampling points is proposed in [23]. It can maintain the ratio of the fictitious sampling frequency to the grid frequency an integer. However, the fictitious sampling points are calculated from practical sampling points by linear calculation, and there are errors between them.

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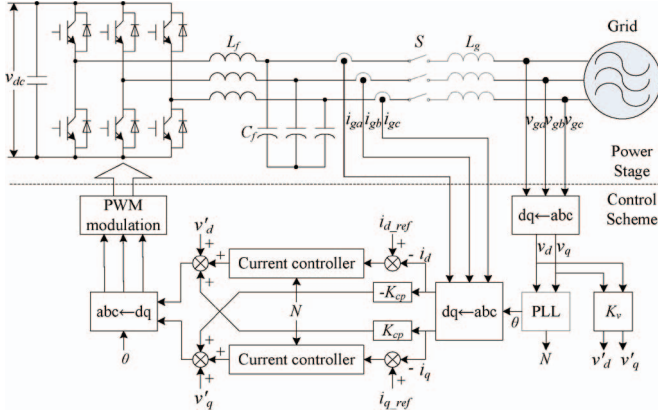


Fig. 1. Block diagram of a three-phase grid-connected inverter with the improved repetitive control scheme.

To solve the issues mentioned earlier, an improved repetitive control scheme is proposed in this paper. It introduces a special designed finite impulse response (FIR) filter, cascaded with a traditional delay function. The FIR filter replaces the auxiliary function for stabilization, and it has adjustable linear characteristic, which means that its delay time is linear to the frequency, and can be adjusted. The proposed control function can approximate the ideal repetitive control function of any ratio. The improved repetitive control scheme varies the FIR filter according to the varied grid frequency and can maintain its resonant points matching the grid fundamental and harmonic frequencies well. Therefore, the grid-connected VSI using this scheme will have low tracking error and THD.

The overall structure of the system is described in Section II. In Section III, the control of the grid-connected VSI is discussed, and the improved repetitive control scheme is analyzed. In Section IV, the simulation and experimental results are presented, which verify the effectiveness of the proposed scheme.

II. DESCRIPTION OF THE SYSTEM

The proposed control system is shown in Fig. 1. The current controller includes the improved repetitive control scheme, so that i_d and i_q could track the references i_{d_ref} and i_{q_ref} , where i_d and i_q are transformed from the grid currents i_{ga} , i_{gb} , and i_{gc} . The real and reactive powers injected into the grid are determined by i_{d_ref} and i_{q_ref} and the grid voltages, while the tracking performance and THD of the grid currents are determined by the current controller. A phase-locked loop (PLL) is used to provide the phase information θ of the grid voltages for dq/abc transformation and the ratio N of the sampling frequency f_s to the grid frequency f_0 for the current controller. When the references i_{d_ref} and i_{q_ref} are equal to zero, the generated voltages should be equal to the grid voltages; then, the circuit breakers could be closed at any time if needed. In order to achieve this, the grid voltages v_{ga} , v_{gb} , and v_{gc} are transformed to v_d and v_q , and they are feedforwarded and added to the output of the current controller via a proportional regulator K_v . i_d and i_q via $\pm K_{cp}$ are also added to the output of the current controller to decouple each other [24]. The inverter is assumed to be powered by a constant dc voltage source; hence, no controller is needed to regulate the dc-link voltage.

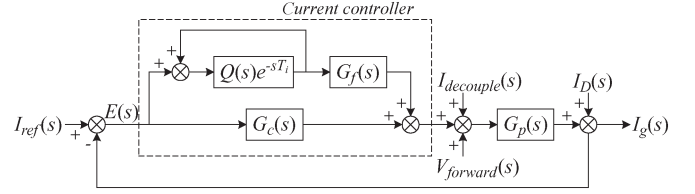


Fig. 2. Block diagram of the improved repetitive control scheme.

In this paper, the improved repetitive control scheme is discussed in a three-phase inverter system in the SRF. The system gain at the fundamental frequency is theoretically infinite with the PI regulator in the SRF, and the repetitive control is used to compensate the harmonics. Certainly, the scheme can also be used in a single-phase system or in the stationary reference frame.

III. DESIGN OF THE IMPROVED REPETITIVE CURRENT CONTROLLER

In this section, the model of the three-phase grid-connected VSI in the SRF is established. Then, the current controller with the PI control and repetitive control techniques is designed based on the model. The block diagram of the improved repetitive control scheme is shown in Fig. 2. $G_p(s)$ is the transfer function of the VSI in the SRF; $G_c(s)$ is the transfer function of the conventional PI control for fundamental current regulation. $Q(s)$, e^{-sT_i} , and $G_f(s)$ are the three factors of the proposed repetitive control scheme. $I_{decouple}(s)$ is the current-decoupling factor, and $V_{forward}(s)$ is the grid voltage feedforward factor. $I_g(s)$ is the VSI output current injected into the grid, $I_{ref}(s)$ is its reference, and $I_D(s)$ is the disturbance of the grid current. The repetitive control scheme proposed in this paper is a plug-in type. Moreover, different from that in the conventional repetitive control technique, the auxiliary function $Q(s)$ in it is replaced by a special designed FIR filter.

A. Model of the Three-Phase Grid-Connected VSI in the SRF

The power stage consists of an insulated gate bipolar transistor (IGBT) inverter bridge, an LC filter (L_f and C_f), and a grid interface inductor L_g , as shown in Fig. 1. The LC filter and the grid interface inductor form an LCL filter. The currents of the two inductors i_L and i_g and the voltage of the capacitor v_C are chosen as state variables; then, the model of the VSI shown in Fig. 1 is given as

$$\frac{d}{dt} \begin{bmatrix} i_{Lab}(t) \\ i_{Lbc}(t) \\ i_{Lca}(t) \end{bmatrix} = -\frac{R_L + R_C}{L_f} \times \begin{bmatrix} i_{Lab}(t) \\ i_{Lbc}(t) \\ i_{Lca}(t) \end{bmatrix} + \frac{R_C}{L_f} \times \begin{bmatrix} i_{gab}(t) \\ i_{gbc}(t) \\ i_{gca}(t) \end{bmatrix} - \frac{1}{L_f} \times \begin{bmatrix} v_{Cab}(t) \\ v_{Cbc}(t) \\ v_{Cca}(t) \end{bmatrix} + \frac{v_{dc}}{L_f} \times \begin{bmatrix} d_{ab}(t) \\ d_{bc}(t) \\ d_{ca}(t) \end{bmatrix} \quad (1)$$

$$\frac{d}{dt} \begin{bmatrix} i_{gab}(t) \\ i_{gbc}(t) \\ i_{gca}(t) \end{bmatrix} = \frac{R_C}{L_g} \times \begin{bmatrix} i_{Lab}(t) \\ i_{Lbc}(t) \\ i_{Lca}(t) \end{bmatrix} - \frac{R_C + R_g}{L_g} \times \begin{bmatrix} i_{gab}(t) \\ i_{gbc}(t) \\ i_{gca}(t) \end{bmatrix} + \frac{1}{L_g} \times \begin{bmatrix} v_{Cab}(t) \\ v_{Cbc}(t) \\ v_{Cca}(t) \end{bmatrix} - \frac{1}{L_g} \times \begin{bmatrix} v_{gab}(t) \\ v_{gbc}(t) \\ v_{gca}(t) \end{bmatrix} \quad (2)$$

$$\frac{d}{dt} \begin{bmatrix} v_{Cab}(t) \\ v_{Cbc}(t) \\ v_{Cca}(t) \end{bmatrix} = \frac{1}{C_f} \times \begin{bmatrix} i_{Lab}(t) \\ i_{Lbc}(t) \\ i_{Lca}(t) \end{bmatrix} - \frac{1}{C_f} \times \begin{bmatrix} i_{gab}(t) \\ i_{gbc}(t) \\ i_{gca}(t) \end{bmatrix} \quad (3)$$

where R_L , R_C , and R_g are the parasitic resistances of L_f , C_f , and L_g . Moreover, d_{ab} , d_{bc} , and d_{ca} are the control signals. In (1)–(3), i_{Lab} means i_{La} minus i_{Lb} , i_{gbc} means i_{gb} minus i_{gc} , etc. Here, the “phase-to-phase” currents and voltages are used instead of the line currents and phase voltages. Then, the uncertain neutral point in the VSI is avoided in the model, while the formulations are suitable for the following abc/dq transformation [25], [26].

The transformation matrix from the stationary reference frame to the SRF is

$$T = \sqrt{\frac{2}{3}} \begin{bmatrix} \sin \omega_0 t & \sin(\omega_0 t - \frac{2\pi}{3}) & \sin(\omega_0 t + \frac{2\pi}{3}) \\ -\cos \omega_0 t & -\cos(\omega_0 t - \frac{2\pi}{3}) & -\cos(\omega_0 t + \frac{2\pi}{3}) \end{bmatrix} \quad (4)$$

with $\omega_0 = 2\pi f_0$, where f_0 is the fundamental frequency of the grid voltages. By multiplying matrix T^{-1} on both sides of (1)–(3), they will be

$$\begin{aligned} \frac{d}{dt} \begin{bmatrix} i_{Ld}(t) \\ i_{Lq}(t) \end{bmatrix} &= -\frac{R_L + R_C}{L_f} \times \begin{bmatrix} i_{Ld}(t) \\ i_{Lq}(t) \end{bmatrix} + \frac{R_C}{L_f} \times \begin{bmatrix} i_{gd}(t) \\ i_{gq}(t) \end{bmatrix} \\ &\quad - \frac{1}{L_f} \times \begin{bmatrix} v_{Cd}(t) \\ v_{Cq}(t) \end{bmatrix} + \frac{v_{dc}}{L_f} \times \begin{bmatrix} d_d(t) \\ d_q(t) \end{bmatrix} - X \times \begin{bmatrix} i_{Ld}(t) \\ i_{Lq}(t) \end{bmatrix} \end{aligned} \quad (5)$$

$$\begin{aligned} \frac{d}{dt} \begin{bmatrix} i_{gd}(t) \\ i_{gq}(t) \end{bmatrix} &= \frac{R_C}{L_g} \times \begin{bmatrix} i_{Ld}(t) \\ i_{Lq}(t) \end{bmatrix} - \frac{R_C + R_g}{L_g} \times \begin{bmatrix} i_{gd}(t) \\ i_{gq}(t) \end{bmatrix} \\ &\quad + \frac{1}{L_g} \times \begin{bmatrix} v_{Cd}(t) \\ v_{Cq}(t) \end{bmatrix} - \frac{1}{L_g} \times \begin{bmatrix} v_{gd}(t) \\ v_{gq}(t) \end{bmatrix} - X \times \begin{bmatrix} i_{gd}(t) \\ i_{gq}(t) \end{bmatrix} \end{aligned} \quad (6)$$

$$\frac{d}{dt} \begin{bmatrix} v_{Cd}(t) \\ v_{Cq}(t) \end{bmatrix} = \frac{1}{C_f} \times \begin{bmatrix} i_{Ld}(t) \\ i_{Lq}(t) \end{bmatrix} - \frac{1}{C_f} \times \begin{bmatrix} i_{gd}(t) \\ i_{gq}(t) \end{bmatrix} - X \times \begin{bmatrix} v_{Cd}(t) \\ v_{Cq}(t) \end{bmatrix} \quad (7)$$

where

$$X = T(dT^{-1}/dt) = \begin{bmatrix} 0 & \omega_0 \\ -\omega_0 & 0 \end{bmatrix}.$$

If $G_p(s)$ is directly derived from (5)–(7), this transfer function will be complex and contain some terms whose coefficients are very small. Moreover, $\omega_0 C_f$ is contained in most of the terms with small coefficients, while the coupling coefficient of the capacitor voltages in (7) is also $\omega_0 C_f$. In reality, the system is physically nonideal; a completely accurate model could not be established. Therefore, in this paper, the coupling of the

capacitor voltages is neglected, which means that the matrix X in (7) is zero. Then, taking the Laplace transformation of (5)–(7) and neglecting the term whose coefficient is $\omega_0^2 L_f C_f L_g$, the transfer function of the VSI in the SRF is finally given in (8), shown at the bottom of the page, where $G_{dp}(s)$ and $G_{qp}(s)$ are the d - and q -axis transfer functions in the SRF. Furthermore, the decoupling coefficient of the current is

$$K_{cp} = \frac{\omega_0(L_f + L_g)}{v_{dc}}. \quad (9)$$

B. Repetitive Controller Design

From Fig. 2, the tracking error of the VSI without the repetitive controller is

$$E_0(s) = \frac{I_{ref}(s) - I_D(s)}{1 + G_c(s)G_p(s)}. \quad (10)$$

According to the internal model principle [9], the steady-state tracking error will be zero only if the models of $I_{ref}(s)$ and $I_D(s)$ are included in the model of $G_c(s)$. Because the reference $I_{ref}(s)$ is a dc value in the SRF, its model can be included in a PI controller. The grid current disturbance $I_D(s)$ at the harmonic frequencies is ac variable, and its model can be included in a repetitive controller. With the repetitive controller in Fig. 2, (10) can be revised as

$$E(s) = E_0(s) \times \frac{1 - Q(s)e^{-sT_i}}{1 - [1 - G_f(s)H(s)]Q(s)e^{-sT_i}} \quad (11)$$

where $H(s) = G_p(s)/(1 + G_c(s)G_p(s))$.

Assuming that the system is stable, the following condition is satisfied:

$$|[1 - G_f(s)H(s)]Q(s)| < 1, \quad s = j\omega \quad (12)$$

then, (11) can be expressed by the Taylor progression as

$$\begin{aligned} E(s) &= E_0(s) \times [1 - Q(s)e^{-sT_i}] \\ &\quad \times \left\{ 1 + [1 - G_f(s)H(s)]Q(s)e^{-sT_i} \right. \\ &\quad \left. + [1 - G_f(s)H(s)]^2 Q^2(s)e^{-2sT_i} + \dots \right\} \\ &= E_0(s) \times \left\{ 1 - G_f(s)H(s)Q(s)e^{-sT_i} \right. \\ &\quad - G_f(s)H(s)[1 - G_f(s)H(s)]Q^2(s)e^{-2sT_i} \\ &\quad - G_f(s)H(s)[1 - G_f(s)H(s)]^2 Q^3(s)e^{-3sT_i} \\ &\quad \left. + \dots \right\} \end{aligned} \quad (13)$$

$$G_p(s) = G_{dp}(s) = G_{qp}(s)$$

$$= \frac{v_{dc}(1 + sR_C C_f)}{L_f L_g C_f s^3 + C_f [L_f(R_g + R_C) + L_g(R_L + R_C)] s^2 + [L_f + L_g + C_f(R_L R_C + R_L R_g + R_C R_g)] s + R_L + R_g} \quad (8)$$

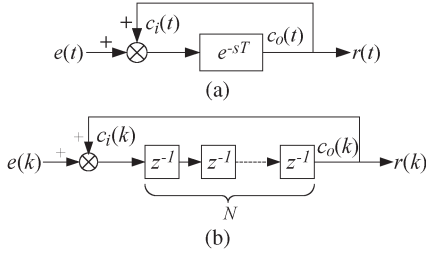


Fig. 3. Block diagram of the repetitive control technique. (a) Time domain. (b) Discrete-time domain.

which indicates that the repetitive controller does not take effect on the first cycle. If

$$1 - G_f(s)H(s) = 0 \quad (14)$$

then the error can be simplified to $E(s) = E_0(s) \times [1 - Q(s)e^{-sT_i}]$ from the second cycle onward.

Meanwhile, if

$$1 - Q(jk\omega_0)e^{-jk\omega_0T_i} = 0, \quad k = 1, 2, 3, \dots \quad (15)$$

the tracking error of (11) can be zero at fundamental and all harmonic frequencies. However, due to the tolerance of the devices, (14) and (15) cannot be practically satisfied. $Q(s)$ is chosen as a low-pass filter or a constant value less than a unit, while $G_f(s)$ is chosen as

$$G_f(s) = \frac{1}{H(s)} \times \frac{p}{s + p} \quad (16)$$

where p is an additional pole above the system crossover frequency for robustness. Then, the harmonic elimination requirements of (14) and (15) would be approximately satisfied, and the stability condition of (12) would be satisfied too.

C. Improved Repetitive Control Scheme With Frequency-Adaptive Capability

The block diagram of the repetitive control technique is shown in Fig. 3. Fig. 3(a) shows the time-domain block diagram, and Fig. 3(b) shows the discrete-time-domain block diagram, which consists of N first-in–first-out (FIFO) cells, where $N = f_s/f_0 = T_0/T_s$ is the ratio of the sampling frequency to the grid frequency. When the grid frequency varies, which is quite common in the DPGS, the ratio cannot maintain an integer. However, only integral FIFO cells could be implemented in the discrete-time domain. This will cause the resonant frequencies of the repetitive control technique to deviate from the real grid fundamental and harmonic frequencies, and the performance of the system will be degraded significantly.

Meanwhile, (15) indicates the following: When the auxiliary function $Q(s)$ is a constant value which is slightly less than a unit, the steady-state tracking error will be a little larger than zero; when $Q(s)$ is a normal low-pass filter, the repetitive control function will be different from the ideal one, although it has unit gain at low frequency. That means that $Q(s)$ will

degrade the system performance, even when the ratio is a fixed integer.

In Fig. 2, if the function $Q(s) \times e^{-sT_i}$ can approach the ideal delay function e^{-sT_0} , the performance of the system will be the same as that using the ideal repetitive control, and the tracking error and THD of the VSI will be minimized. Letting N_i be an integer smaller than N and $T_i = N_i T_s$, then it can be revised as that $Q(s)$ should approach the function $e^{-s(N-N_i)T_s}$. Let $D = N - N_i$, and the frequency characteristic of the delay function e^{-sDT_s} is

$$\angle e^{-sDT_s} = \angle e^{-j\omega DT_s} = -D\omega T_s. \quad (17)$$

It is a linear-phase characteristic. This demands that $Q(s)$ also has linear-phase characteristic. Moreover, $Q(s)$ should be a low-pass filter which satisfies (15). In this paper, the FIR digital filter is proposed to replace the auxiliary function $Q(s)$, because it can satisfy all the aforementioned requirements. Its normal transfer function in the discrete-time domain is

$$F(z) = \sum_{i=0}^{M-1} a_i z^{-i} \quad (18)$$

where M is the order of the FIR filter.

The traditional design methods for FIR filters include the window function method, the frequency sampling method, etc. However, the frequency characteristics of these FIR filters are determined by the order M , and they can only be $-0.5\omega T_s, -\omega T_s, -1.5\omega T_s, -2\omega T_s, \dots$ [27]. Here, a new design methodology for the FIR filter is proposed as follows. The transfer function of the proposed FIR filter is

$$Q(z) = \sum_{k=0}^{M-1} q(k) z^{-k}. \quad (19)$$

Moreover, the relationship of its coefficients is given as

$$q(M-1-k) = \frac{D-k}{(M-1-k)-D} q(k) \quad (0 \leq k \leq M-1) \quad (20)$$

where the coefficients $q(k)$ ($0 \leq k \leq (M-1)/2$) are designed by the rectangle window function method and D is chosen as $(M-3)/2 < D \leq (M-1)/2$.

A FIR filter with the known frequency characteristic of (17) can be designed simply based on (19) and (20). The principle of it is analyzed in the following.

1) *Frequency Characteristic*: The frequency characteristic of (19) is

$$\begin{aligned} \angle Q(z) &= \angle Q(e^{j\omega T_s}) \\ &= \angle \left\{ \sum_{k=0}^{M-1} q(k) [\cos(k\omega T_s) - j \sin(k\omega T_s)] \right\} \\ &= \arctan \frac{-\sum_{k=0}^{M-1} q(k) \sin(k\omega T_s)}{\sum_{k=0}^{M-1} q(k) \cos(k\omega T_s)}. \end{aligned} \quad (21)$$

If (21) is equal to (17), then

$$\begin{aligned} & -\sum_{k=0}^{M-1} q(k) \sin(k\omega T_s) \\ & \frac{\sum_{k=0}^{M-1} q(k) \cos(k\omega T_s)}{\sum_{k=0}^{M-1} q(k) \cos(k\omega T_s)} = \tan(-D\omega T_s) \\ & = \frac{-\sin(D\omega T_s)}{\cos(D\omega T_s)} \\ & \Rightarrow \sum_{k=0}^{M-1} q(k) \sin[(D-k)\omega T_s] = 0. \quad (22) \end{aligned}$$

At low frequency, the fundamental and main harmonic frequencies are far smaller than the sampling frequency f_s ; then

$$\sin[(D-k)\omega T_s] = (D-k) \times \sin(\omega T_s). \quad (23)$$

When M is an even number, taking (20) and (23) into (22), (22) can be proved to be true finally. When M is an odd number, there is a middle term of $k = (M-1)/2$ in (22), but it is very small and can be neglected; then, (22) can be proved to be true too. That means that the FIR filter of (19) and (20) can approach the delay function e^{-sDT_s} at low frequency, and then, the improved repetitive control scheme with it can approximate the ideal repetitive control.

2) *Passband, Transition Band, and the Choice of M* : The relationship of the FIR filter's coefficients is determined by (20), while the concrete coefficients $q(k)$ ($0 \leq k \leq (M-1)/2$) still need to be designed. Here, the traditional methods can be used. It is because that, although only the first half part of its coefficients are designed by the traditional methods, the passband and transition band of the proposed FIR filter are similar to the traditional ones.

The corner frequency f_{corner} of the FIR filter should be smaller than one-tenth of the switching frequency f_{switch} for system stabilization. In a digital system, the ratio n of the switching frequency to the sampling frequency is a positive integer; then

$$2\pi f_{\text{corner}} T_s \leq 2\pi \frac{f_{\text{switch}}}{10} T_s = 2\pi \frac{f_s/n}{10} T_s = \frac{\pi}{5n} \quad (24)$$

where the allowed maximum passband is $\pi/5$ when the sampling frequency is equal to the switching frequency.

To achieve good performance, the transition band of the FIR filter should be as small as possible. In this paper, the rectangle window function method is adopted because it has the minimum transition band among the traditional FIR filter design methods [27]. The transition band BW_t using the rectangle window function method is

$$BW_t = \frac{1.8\pi}{M}. \quad (25)$$

The order M of the proposed FIR filter can be determined by (25). In (25), when BW_t is smaller, the order M will be higher. Moreover, if M is too high, the calculation of the coefficients of the FIR filter will be complex and time consuming when the grid frequency varies. Therefore, there is a tradeoff between the performance and the complexity when choosing M .

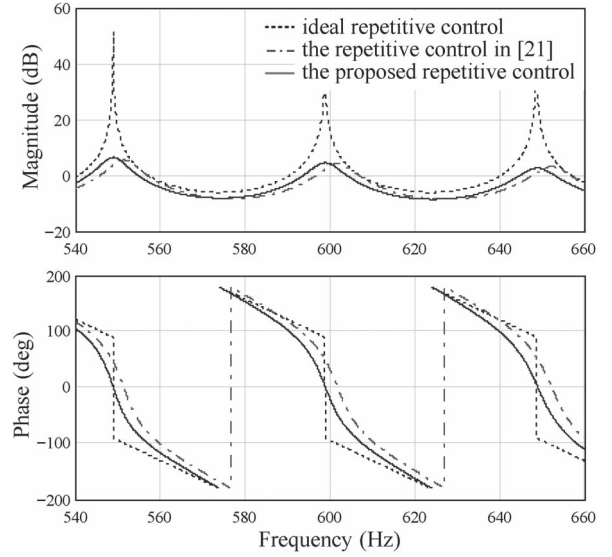


Fig. 4. Bode plot of different repetitive control functions.

3) *Choice of D* : Notice that, when D is $(M-1)/2$, (20) will be revised as

$$q(M-1-k) = q(k) \quad (0 \leq k \leq M-1) \quad (26)$$

which is the same with the traditional FIR filter design methods. It means that D should be chosen around $(M-1)/2$ according to the varied grid frequency. Meanwhile, the value of N_i should be slightly less than the ratio N [28], which demands that the value of D should be slightly larger than zero, and the order M is not too high. Therefore, D is chosen as $(M-3)/2 < D \leq (M-1)/2$ in this paper.

Finally, the design process of the proposed FIR filter is summarized as follows. First, the order M is determined from (25). Then, D is chosen between $(M-3)/2$ and $(M-1)/2$ according to the varied grid frequency. The coefficients $q(k)$ ($0 \leq k \leq (M-1)/2$) are designed based on (24) using the rectangle window function method. Finally, the coefficients $q(k)$ $[(M-1)/2 < k \leq (M-1)]$ are calculated with (20).

Therefore, with the proposed design method mentioned earlier, a FIR filter could be designed quickly and accurately at different ratios N , which has linear-phase low-pass characteristics, and can approach the delay function e^{-sDT_s} at low frequency. The improved repetitive control scheme with this FIR filter can approximate the ideal repetitive control technique and maintain its resonant frequencies matching the grid fundamental and harmonic ones, no matter if the ratio is a fixed integer, a noninteger, or a variable value according to the varied grid frequency. Therefore, the VSI system could have excellent tracking performance and high harmonic compensation with this scheme.

Fig. 4 shows the Bode plot of the functions of the ideal repetitive control, the repetitive control mentioned in [21], and the proposed repetitive control between 540 to 660 Hz. Here, the sampling frequency f_s is 10 kHz, and the grid frequency is 49.9 Hz; then, $N = 200.4$. In this paper, the transition band BW_t of the FIR filter is chosen to be $\pi/5$; then, the order of the

TABLE I
PARAMETERS OF THE VSI SYSTEM

Symbol	Quantity	Value
L_f	filter inductor	600 μ H
C_f	filter capacitor	40 μ F
L_g	grid interface inductor	40 μ H
R_L	parasitic resistance of filter inductor	0.01 Ω
R_C	parasitic resistance of filter capacitor	0.003 Ω
R_g	parasitic resistance of grid interface inductor	0.001 Ω
v_g	grid voltage	380V
v_{dc}	DC bus voltage	720V
P_o	apparent power	30kW

FIR filter is $M = 9$ from (25). Then, D is chosen as $D = 3.4$, while $N_i = 197$. From Fig. 4, compared to the method mentioned in [21], the improved repetitive control scheme could approximate the characteristics of the ideal repetitive control well at the harmonic frequencies.

D. PLL Design

In the improved repetitive control scheme, the coefficients of the proposed FIR filter are calculated online according to the real grid frequency, which is provided by the PLL. Therefore, the performance of the PLL will influence this control. If the output error of the PLL is large, the system performance with the proposed scheme will be degraded by the large frequency error. Thus, the output error of the PLL should be as small as possible.

In this paper, the method using abc/dq transformation of the grid voltages is adopted to design the PLL [29], and the output of the PLL is calculated at the sampling frequency. The bandwidth of the PI regulator in the PLL is designed narrow; then, the output error of the PLL will be small [29]. Although the transient response of the PLL will be slow, it is not an issue since the repetitive control itself is also very slow.

Meanwhile, when the PLL is calculated at the sampling frequency, for example, 10 kHz, the coefficients of the proposed FIR filter can be calculated at 10 kHz, 5 kHz, etc. In this paper, a nine-order FIR filter is chosen, which means that only nine coefficients need to be calculated online. It will just take a little time of the MCU, which is about 3 μ s for the 40-MHz DSP. Therefore, in order to simplify the design process, the coefficients of the proposed FIR filter are calculated at each sampling period in this paper finally.

IV. SIMULATION AND EXPERIMENTAL VALIDATION

A. Controller Design

The parameters of the system are given in Table I. The rated grid frequency is 50 Hz. The PWM switching frequency and the sampling frequency are both 10 kHz. Taking the values in Table I into (8), the transfer function of the VSI is

$$G_p(s) \approx \frac{9 \times 10^7 (s + 8.333 \times 10^6)}{(s + 17.187)(s^2 + 104.479s + 6.667 \times 10^8)}. \quad (27)$$

Moreover, the PI controller is chosen as

$$G_c(s) = 0.0036 + \frac{3.3912}{s}. \quad (28)$$

Then

$$H(s) = \frac{G_p(s)}{1 + G_c(s)G_p(s)} \approx \frac{9 \times 10^7 s(s + 8.333 \times 10^6)}{(s + 1474)(s + 2542)(s^2 - 3893.92s + 6.789 \times 10^8)}. \quad (29)$$

For simplicity and adopting the low-pass characteristic of $H(s)$, the regulator $G_f(s)$ is finally chosen as

$$G_f(s) = \frac{(s + 1474)(s + 2542)}{1 \ 104 \ 680 \times s} \times \frac{10 \ 000}{s + 10 \ 000}. \quad (30)$$

Then, (14) could be satisfied at low frequency.

The nine-order FIR filter of the improved repetitive control scheme should be varied according to the grid frequency variation. For example, if the grid frequency is 49.9 Hz, then $N = 200.4$, $D = 3.4$, and $N_i = 197$. When the rectangle window function method is implemented to design the coefficients $q(k)$ ($0 \leq k \leq 4$), the corner frequency f_{corner} is chosen to be 400 Hz. Then, the coefficients $q(k)$ ($4 < k \leq 8$) are determined from (20). The FIR filter at the 49.9-Hz grid frequency is shown as

$$Q(z) = 0.12108 + 0.13088z^{-1} + 0.13816z^{-2} + 0.14265z^{-3} + 0.1802z^{-4} + 0.03566z^{-5} + 0.07439z^{-6} + 0.08725z^{-7} + 0.08949z^{-8}. \quad (31)$$

The output of the PLL and the coefficients of the proposed FIR filter are both calculated at 10 kHz. Then, the PI regulator in the PLL is designed slower than the conventional one [29], which is finally chosen as

$$0.2338 + \frac{10.39}{s}. \quad (32)$$

B. Simulation Results

Simulations using Simulink in MATLAB software are carried out. The maximal step size is 500 ns, and the sampling time is 100 μ s. The rated output power is 30 kW, and the output current of each phase is 45 A at rms.

In the SRF, the fundamental currents can be well controlled with the PI regulator, and the repetitive control is used to compensate the harmonics. In order to verify the effectiveness of the proposed repetitive control scheme, the simulation results with the conventional repetitive control are also presented for comparison. The output current and the current tracking error at 49.9-, 50-, and 50.1-Hz grid frequencies are shown in Figs. 5–7, respectively. At the 49.9-Hz grid frequency, the output current THD is 2.53% with the improved repetitive control, compared to 3.84% with the conventional one; at 50 Hz, they are 2.51%

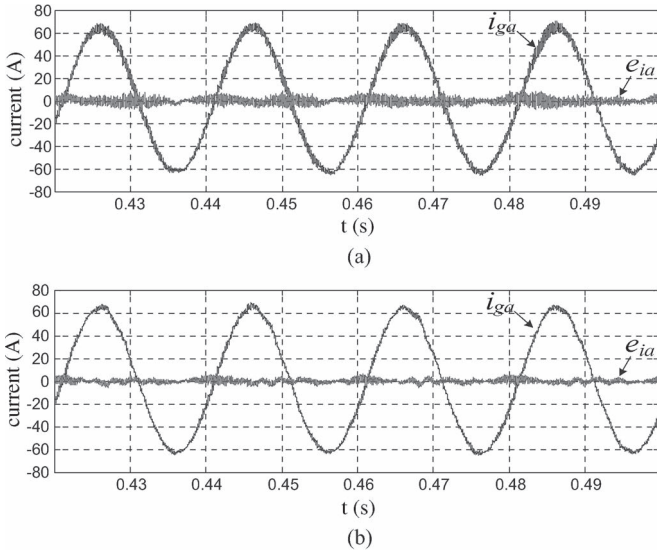


Fig. 5. Simulation results at the 49.9-Hz grid frequency. Output current i_{ga} and current tracking error e_{ia} . (a) With the conventional repetitive control. (b) With the improved repetitive control.

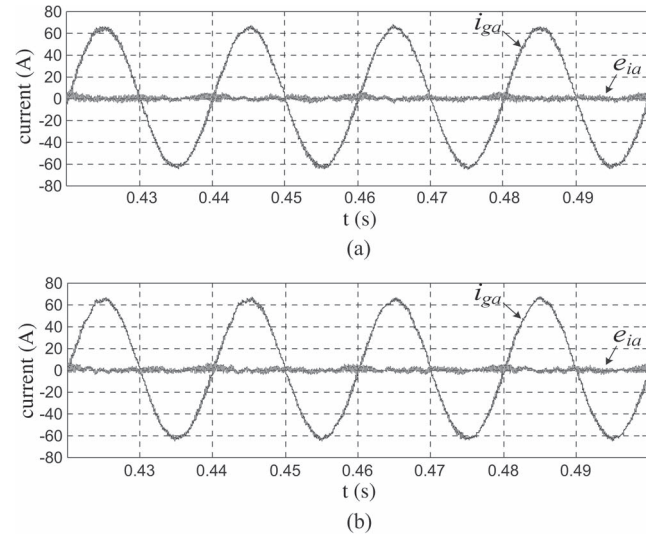


Fig. 6. Simulation results at the 50-Hz grid frequency. Output current i_{ga} and current tracking error e_{ia} . (a) With the conventional repetitive control. (b) With the improved repetitive control.

and 3.02%, and at 50.1 Hz, they are 2.56% and 3.47%. The performance with the conventional repetitive control is depended on the grid frequency, while the improved repetitive control scheme keeps the certain good system performance, which is better than that with the conventional one. The simulation results indicate the effect of the proposed scheme on reducing the steady-state tracking error and compensating the harmonics of the system.

C. Experimental Results

The experimental setup is shown in Fig. 8. The prototype consists of dc bus capacitors, an IGBT inverter bridge, a three-phase LC filter, a three-phase grid interface inductor, voltage and current sensors, and a DSP control board. The IPM PM300CLA120 from Mitsubishi is used for the IGBT inverter

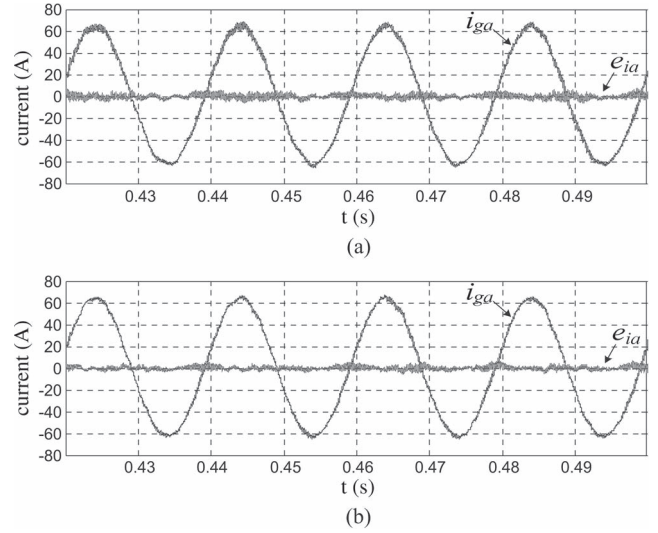


Fig. 7. Simulation results at the 50.1-Hz grid frequency. Output current i_{ga} and current tracking error e_{ia} . (a) With the conventional repetitive control. (b) With the improved repetitive control.

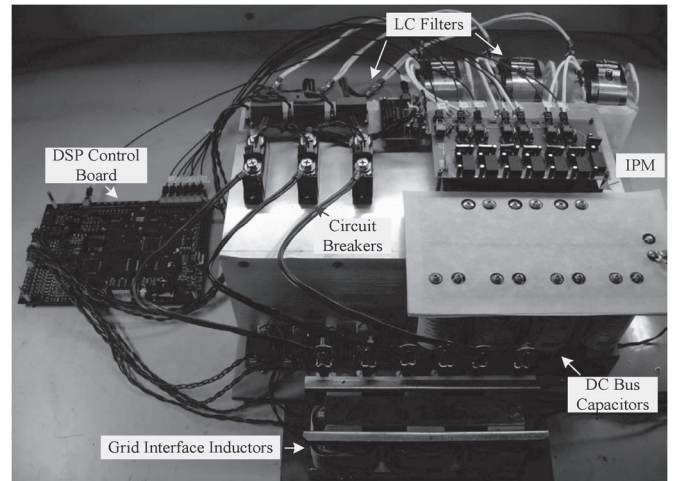


Fig. 8. Experimental setup.

bridge. The generated three-phase voltages by the inverter are connected to the grid via controlled circuit breakers for safety. The rated output power is also 30 kW. In the experiments, the FIR filter of the improved repetitive control scheme is varied according to the grid frequency. The method with a variable one-order low-pass filter in [21] is also presented for comparison, in order to validate the effectiveness of the proposed scheme in practical application.

1) *Under the Balanced System:* The current reference i_{d_ref} is set at 1.732×45 A, and i_{q_ref} is set at zero. This corresponds to a unity power factor, and the current injected into the grid of each phase is 45 A at rms.

Fig. 9(a) and (b) shows the output current i_{ga} , the current tracking error e_{ia} , and the spectrums of i_{ga} with the two control methods. It is clear that the current tracking error with the improved repetitive control scheme is smaller than that with the repetitive control method given in [21]. Also, with the improved repetitive control scheme, the THD is reduced from 4.14% to

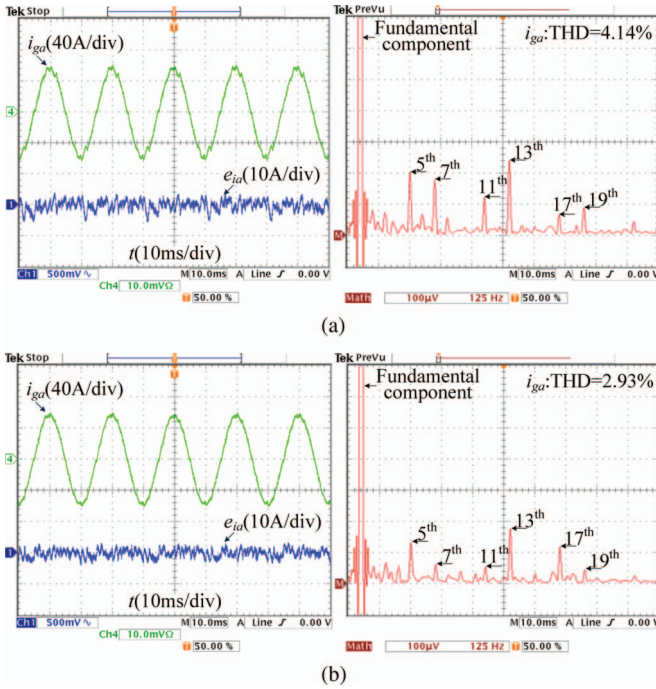


Fig. 9. Experimental results under the balanced system. (Left) Output current i_{ga} and current tracking error e_{ia} . (Right) Spectrums of i_{ga} . (a) With the repetitive control in [21]. (b) With the proposed repetitive control.

2.93%. The experimental results show that the proposed scheme is effective under the balanced system.

2) *Under the Unbalanced System:* In these experiments, two different local loads are connected to the VSI. In the first case, two 2-kW resistive local loads are connected to phases A and B of the VSI, while no local load is connected to phase C. In the second case, a 6-kW three-phase uncontrolled rectifier load is connected to the VSI as the balanced nonlinear local load. The current reference i_{d_ref} is set at 1.732×36 A, and i_{q_ref} is set at zero. Then, the current injected into the grid of each phase is 36 A at rms.

Fig. 10 shows the experimental results with the unbalanced resistive local loads. In Fig. 10, the current tracking error with the proposed scheme is much smaller than that with the method given in [21]. Moreover, the THD using the proposed scheme is 3.28% as shown in Fig. 10(b), which is much lower than that with the repetitive control method proposed in [21], i.e., 5.09% as shown in Fig. 10(a).

Fig. 11 shows the experimental results with a balanced nonlinear local load. Compared with the method given in [21], the proposed scheme has much lower tracking error, and the THD is reduced from 7.69% to 3.51%.

Both the experimental results indicate that the improved repetitive control scheme can effectively eliminate the steady-state tracking error and reduce the current THD under the unbalanced systems.

3) *With Nonintegral Frequency Ratio:* It is hard to change the grid frequency artificially in practice; therefore, in this paper, the sampling frequency is changed to simulate the grid frequency variation. In these experiments, the switching period is set to 99.8 and 100.2 μ s to simulate a lower and a higher grid frequency. Because the sampling frequency is the same with

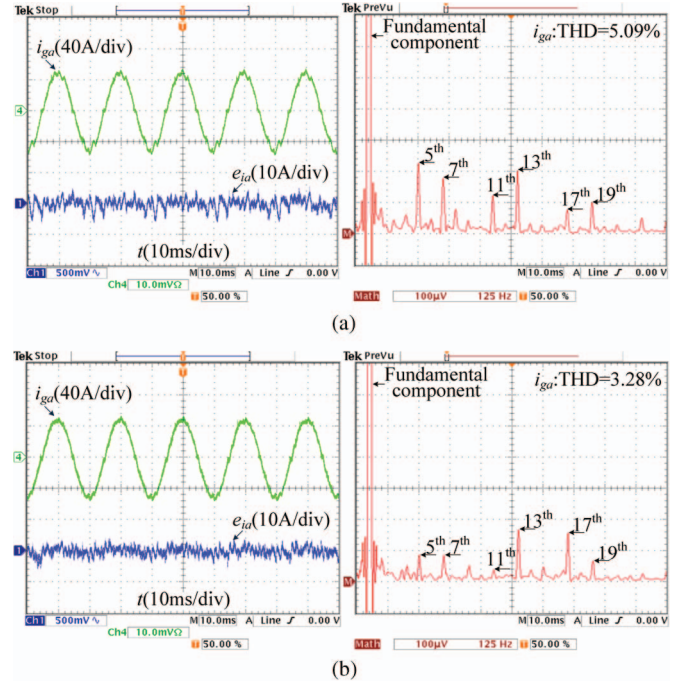


Fig. 10. Experimental results with unbalanced resistive local loads. (Left) Output current i_{ga} and current tracking error e_{ia} . (Right) Spectrums of i_{ga} . (a) With the repetitive control in [21]. (b) With the proposed repetitive control.

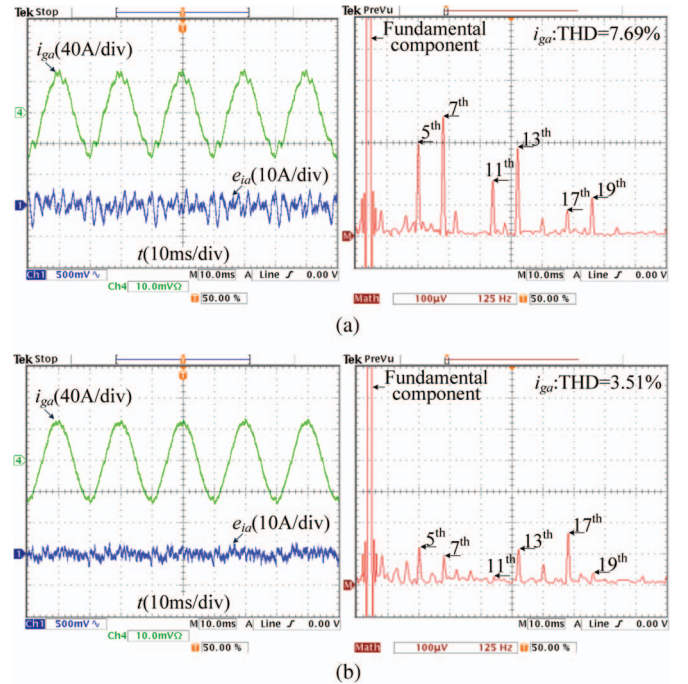


Fig. 11. Experimental results with a balanced nonlinear local load. (Left) Output current i_{ga} and current tracking error e_{ia} . (Right) Spectrums of i_{ga} . (a) With the repetitive control in [21]. (b) With the proposed repetitive control.

the switching frequency, the frequency ratios N are 200.4 and 199.6 with the rated grid frequency, respectively. Also, in these experiments, the FIR filter is changed according to the varied grid frequency as analyzed previously.

The current reference i_{d_ref} is set at 1.732×45 A, and i_{q_ref} is set at zero. The output current i_{ga} , the current tracking error

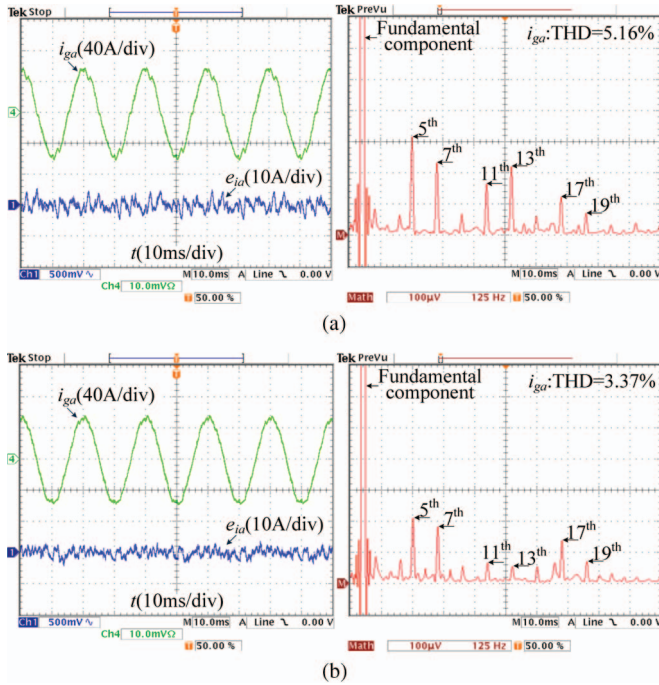


Fig. 12. Experimental results at 99.8- μ s sampling period. (Left) Output current i_{ga} and current tracking error e_{ia} . (Right) Spectrums of i_{ga} . (a) With the repetitive control in [21]. (b) With the proposed repetitive control.

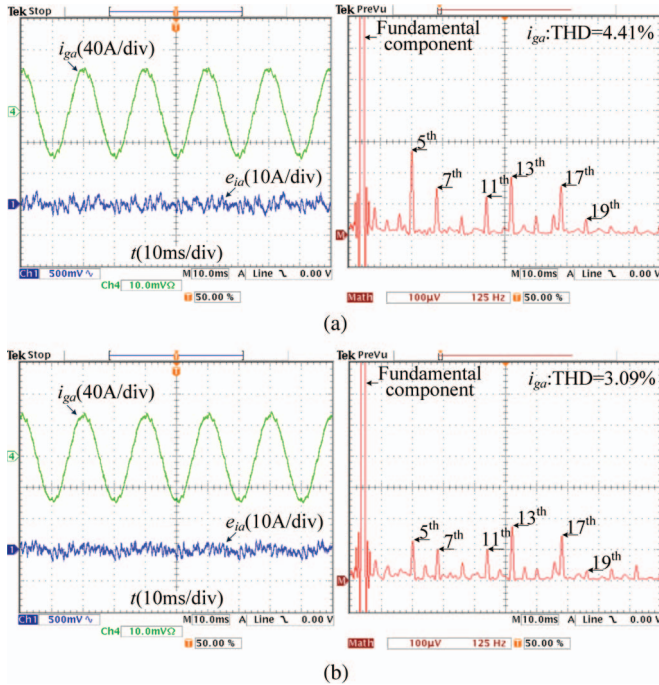


Fig. 13. Experimental results at 100.2- μ s sampling period. (Left) Output current i_{ga} and current tracking error e_{ia} . (Right) Spectrums of i_{ga} . (a) With the repetitive control in [21]. (b) With the proposed repetitive control.

e_{ia} , and the spectrums of i_{ga} are shown in Figs. 12 and 13, respectively.

As shown in Fig. 12, at 99.8- μ s sampling period, the THDs are 3.37% with the proposed scheme and 5.16% with the method given in [21]. At 100.2- μ s sampling period, the THDs are 3.09% with the proposed scheme and 4.41% with the method given in [21], which are shown in Fig. 13.

The experimental results show that the improved repetitive control scheme can keep good system performance with different frequency ratios. It is the same case when the grid frequency varies while the sampling frequency keeps unchanged.

V. CONCLUSION

In this paper, an improved repetitive control scheme has been proposed. It adopts a new FIR filter design method, which has adjustable linear-phase low-pass characteristics. Moreover, the proposed scheme can approximate the ideal repetitive control when the ratio of the sampling frequency to the grid frequency varies. The principle of the control scheme and the FIR filter design method have been analyzed. Through simulations and experiments, it is demonstrated theoretically that the improved repetitive control scheme is effective to improve the tracking performance and reduce the harmonic distortion for the grid-connected VSI systems. Furthermore, it can also be used in islanding-mode inverter and active power filter systems.

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